Intelligent Fall Monitoring System for Elderly Care

INDIVIDUAL THESIS

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Individual component: Environmental Risk Mapping and Smart Safety Zone System

Group ID: R26-IT-051

Report submitted in partial fulfilment of the requirements for the Bachelor of Science (Hons) in Information Technology Specialize in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

2026

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ABSTRACT

Falls are a major cause of injuries and loss of independence among older adults. Many falls are associated with environmental hazards within eldercare environments, such as poor lighting, wet floors, and obstacles that block walking paths. Most existing fall-related technologies mainly focus on detecting falls after they occur or estimating risk based only on the person's physical condition. Environmental risk assessment is still often performed manually using periodic checklists. This research presents the Environmental Risk Mapping & Smart Safety Zone System, an environmental awareness component of the Intelligent Fall Monitoring System for Elderly Care. The main objective of this system is to monitor the surroundings of older adults in real time and transform the collected environmental data into a zone-level risk map to support safer movement and prevent potential falls. The proposed system uses the following hardware components: MPU-6050 HW-123E – Used to detect motion, acceleration, and orientation changes. BH1750 Light Intensity Sensor – Used to measure the amount of light in the environment. VL53L0X ToF Distance Sensor – Used to measure the distance to nearby objects and identify potential obstacles. Moisture Sensor – Used to detect wet-floor conditions that may increase the risk of slipping. ESP32 Microcontroller – Used to collect and process sensor data. ESP32-CAM – Used for camera-based environmental monitoring and hazard detection. Buzzer – Used to provide an audible warning when a configured dangerous condition is detected.

The devices publish discovery, status, and sensor telemetry messages through the MQTT protocol to a HiveMQ Cloud broker. A Python application maintains a thread-safe registry of devices and zones. Devices and zones are automatically discovered and registered based on incoming messages rather than relying on predefined room layouts. An explainable additive risk-scoring mechanism calculates a risk score between 0 and 100 for each sensor node. The maximum allocated scores are 30 points for lighting conditions, 35 points for proximity, 30 points for moisture, and 5 points for motion. The risk level of each zone is determined by the highest risk score among the sensor nodes assigned to that zone. Accordingly, zones are classified into three categories: LOW (low risk), MEDIUM (moderate risk), and HIGH (high risk). Additionally, YOLOv8 is used to analyse camera data and detect potential environmental hazards. An automatic buzzer control mechanism is implemented using moisture thresholds with hysteresis. The buzzer is activated when the moisture level reaches or exceeds 30% and deactivated when the level falls to 20% or below. This approach helps prevent the buzzer from repeatedly switching on and off due to small fluctuations in moisture readings. A dashboard developed using Streamlit displays device health, real-time sensor readings, and a dynamic 3D environmental risk map.

Functional verification was conducted using a prototype installed in a staircase environment. The tests confirmed that the sensor node and camera were automatically discovered and assigned to their corresponding zone. The dashboard refreshed every second, and the displayed risk scores of 55, 60, and 70 were observed to correspond with the defined scoring rules and colour bands. Code review also identified a risk-scoring defect and several security limitations. These issues and their corresponding fixes were documented as part of the system evaluation. This system component introduces an explainable multimodal environmental risk assessment model, a self-configuring zone architecture, and an integrated environmental monitoring dashboard. By identifying potential environmental hazards and representing them through dynamic safety zones, the system provides valuable environmental context to the person-centred components of the overall Intelligent Fall Monitoring System for Elderly Care. It aims to support safer movement and contribute to fall prevention among older adults.

Keywords: Fall Prevention, Environmental Risk Mapping, Internet of Things (IoT), MQTT, YOLOv8, Smart Safety Zones.

ACKNOWLEDGEMENTS

I sincerely thank my supervisor, Mr. Uditha Dharmakeerthi, and co-supervisor, Mr. Kanishka Yapa, for their valuable guidance, constructive feedback, and continuous encouragement throughout this research. I am also grateful to my group members, M. T. K. Chathuri, W. A. H. Divyanjalie, and E. D. K. M. Kulathunga, whose individual components of the Intelligent Fall Monitoring System for Elderly Care made the successful integration of the overall system possible. Furthermore, I would like to express my sincere appreciation to the lecturers who served on the research panel for their valuable suggestions, constructive feedback, and guidance, which helped us improve our research. Finally, I extend my heartfelt thanks to my family and friends for their continuous support, understanding, and encouragement throughout my academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADC	Analogue-to-Digital Converter
AI	Artificial Intelligence
COCO	Common Objects in Context (dataset)
CSV	Comma-Separated Values
FPS	Frames Per Second
GPIO	General-Purpose Input/Output
HTTP	Hypertext Transfer Protocol
I2C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
IoT	Internet of Things
JPEG	Joint Photographic Experts Group (image format)
JSON	JavaScript Object Notation
MJPEG	Motion JPEG
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
PIR	Passive Infrared (sensor)
PSRAM	Pseudo-Static Random-Access Memory
RF	Random Forest
RGB565	16-bit colour pixel format
SDG	Sustainable Development Goal
TLS	Transport Layer Security
ToF	Time of Flight
YOLO	You Only Look Once (object detector)

1. INTRODUCTION

1.1 Overall project context

Intelligent Fall Monitoring System for Elderly Care system for Eldercare is an intelligent, IoT-based healthcare monitoring system designed to reduce the risk of falls among elderly people by combining activity recognition, environmental hazard detection, physiological monitoring, and real-time fall prevention.

Traditional fall detection systems mainly focus on identifying a fall after it occurs. However, this project takes a more proactive approach by continuously monitoring the elderly person's movements, surrounding environmental conditions, and physiological indicators to identify potential fall risks and provide timely interventions.

The proposed system consists of four main components. First, the Context-Aware Activity Recognition component uses an IMU-based wearable device to recognize activities such as walking, standing, and falling while considering the person's movement patterns and stability. Second, the Environmental Risk Mapping and Smart Safety Zone System uses IoT sensors and ESP32-CAM modules to identify environmental hazards, including wet floors, poor lighting, obstacles, and unsafe furniture. It transforms the detected hazards into dynamic safety zones classified according to their risk levels. Third, the BioSignal Predictor monitors physiological indicators such as heart rate and estimates blood pressure to help identify health-related conditions that may contribute to fall risk. Finally, the Real-Time Adaptive Fall Prevention and Smart Alert System integrates the outputs of the other components to determine suitable responses, including voice warnings, caregiver notifications, and emergency alerts. It can also provide appropriate recovery guidance after a fall based on the person's condition and movement.

All four components work together to provide a comprehensive understanding of the elderly person's safety status. By combining information about the person's activities, environmental hazards, and physiological condition, the system aims to identify potential risks early, support timely preventive actions, and improve the safety and independence of elderly people. Overall, the project aims to move beyond conventional fall detection toward a proactive, context-aware, and intelligent fall prevention system that supports continuous elderly care and reduces the likelihood and consequences of fall-related injuries.

1.2 Individual component context

The Environmental Risk Mapping and Smart Safety Zone System is a key component of the overall research project, **Intelligent Fall Monitoring System for Elderly Care** The main objective of this component is to continuously monitor the environments where elderly people live or move, identify potential environmental hazards, and help reduce fall risks before an accident occurs.

Traditional fall prevention methods often rely on periodic environmental inspections or static safety assessments. As a result, changing hazards such as wet floors, insufficient lighting, obstacles on the floor, and unsafe furniture arrangements may not be identified in a timely manner. To address this problem, the proposed component integrates IoT sensors, ESP32-CAM modules, and intelligent data analysis techniques to identify and assess environmental risks.

The system collects environmental data using light sensors, moisture sensors, and other relevant sensors to monitor lighting levels, floor conditions, and distances to nearby obstacles. In addition, the ESP32-CAM modules provide live video streams of the monitored environment. The video streams are transmitted to a PC, where a YOLO-based AI model processes the video frames to detect people and objects**.** The detection results are then displayed on the system dashboard in real time. By combining sensor readings with camera-based detection results, the system can provide a more comprehensive assessment of environmental safety.

Based on the collected sensor data and detected hazards, the system generates a Dynamic Environmental Risk Map that visually represents the risk levels of different areas within the monitored environment. The environment can be classified into the following Smart Safety Zones according to the assessed risk levels:

- Safe Zone: An area where no significant environmental hazards have been detected.
- Caution Zone: An area where hazards such as insufficient lighting or minor obstacles require additional attention.
- Danger Zone: An area where high-risk hazards, such as wet or slippery floors, significant obstacles, or other dangerous conditions, have been identified.

These risk levels can be updated as environmental conditions change. The system can also provide visual warnings and buzzer alerts to notify users about potentially hazardous areas. However, to avoid unnecessary alerts, conditions such as low lighting alone should be

evaluated in context. Alert decisions should consider the severity and duration of the detected condition, together with other available sensor readings.

Furthermore, the environmental risk information generated by this component can be integrated with the other components of the overall fall monitoring system. By combining environmental hazards with information about the elderly person's activities and physiological condition, the overall system can support more informed decisions and appropriate preventive actions.

1.3 Problem statement

Environmental hazards are significant contributing factors to falls among elderly people. Hazards such as wet or slippery floors, insufficient lighting, obstacles, stairs, and unsafe furniture arrangements can increase the risk of falls in residential environments. However, many existing fall prevention approaches rely on periodic safety inspections or focus primarily on detecting a fall after it has occurred, rather than continuously monitoring environmental conditions and identifying potential hazards in real time.

Furthermore, environmental conditions can change throughout the day, making it difficult to identify and communicate emerging risks using static assessments or conventional monitoring methods. The lack of continuous environmental monitoring, real-time hazard detection, and dynamic risk visualization limits the ability of elderly individuals and caregivers to recognize dangerous areas and take preventive action promptly.

Therefore, there is a need for an intelligent environmental monitoring system that integrates IoT sensors and camera-based AI object detection to identify potential hazards, assess environmental risk levels, and represent them through a dynamic environmental risk map with different Smart Safety Zones. Such a system could provide timely visual warnings and audible alerts to help elderly individuals and caregivers recognize hazardous areas, make safer movement decisions, and reduce the likelihood of environment-related falls.

1.4 Research gap

Although existing fall prevention systems use wearable devices, motion sensors, and camera-based technologies to detect falls or monitor elderly people, many approaches primarily focus on the individual's movements or identify a fall after it has occurred. Environmental hazards such as wet floors, insufficient lighting, obstacles, and unsafe furniture arrangements may not be continuously assessed and represented in a unified, real-time risk map.

Furthermore, systems that monitor environmental conditions may rely on individual sensor readings or predefined thresholds without combining multiple environmental indicators with AI-based visual object detection. As a result, identifying changing hazards, understanding their combined risk, and clearly communicating the risk level of each monitored area can remain challenging. In addition, insufficient context-aware alert mechanisms may lead to unnecessary warnings, such as alerts triggered by low lighting alone during nighttime.

Therefore, a research gap exists in developing an integrated environmental monitoring approach that combines IoT sensor data with PC-based, YOLO-driven video analysis to identify environmental hazards, assess their risk levels, and dynamically represent them through Safe, Caution, and Danger zones. The proposed Environmental Risk Mapping and Smart Safety Zone System aims to address this gap by providing continuously updated environmental risk information and context-sensitive warnings to support safer movement and proactive fall prevention for elderly people.

1.5 Aim and objectives

1.5.1 Aim

The main aim of this research component is to develop an intelligent environmental monitoring system that integrates IoT sensor data and AI-based visual object detection to identify environmental hazards, assess environmental risk levels, and dynamically represent them through a risk map with Smart Safety Zones. The system aims to support proactive fall prevention among elderly people by providing real-time environmental risk information and appropriate warnings to help identify potentially hazardous areas.

1.5.2 Objectives

The specific objectives of this research component are:

1. To monitor environmental conditions relevant to fall risks, including lighting levels, floor moisture, and nearby obstacles, using appropriate IoT sensors.
2. To implement ESP32-CAM-based video streaming and PC-based YOLO object detection to identify relevant objects and potential environmental hazards within monitored areas.
3. To integrate sensor readings and camera-based detection results to assess environmental hazards and determine the corresponding risk levels.
4. To develop a Dynamic Environmental Risk Map that visually represents the assessed risk levels of monitored areas using Safe, Caution, and Danger zones.

5. To implement context-sensitive warning mechanisms that provide appropriate visual or audible alerts when significant environmental hazards are detected, while minimizing unnecessary warnings.
6. To evaluate the performance of the proposed system in detecting selected environmental hazards, classifying risk levels, updating the risk map, and generating appropriate warnings under different environmental conditions.

1.6 Research questions or hypotheses

The following research questions are formulated to guide the development and evaluation of the proposed Environmental Risk Mapping and Smart Safety Zone System.

RQ1: How effectively can IoT sensors monitor environmental conditions, such as lighting levels and floor moisture, that may contribute to fall risks among elderly people?

RQ2: To what extent can PC-based YOLO object detection using ESP32-CAM video streams identify relevant objects and potential environmental hazards in monitored areas?

RQ3: How effectively can IoT sensor readings and AI-based visual detection results be integrated to assess environmental risk levels?

RQ4: How accurately can the proposed system classify monitored areas into Safe, Caution, and Danger zones and update the Dynamic Environmental Risk Map when environmental conditions change?

RQ5: How effectively can context-sensitive warning mechanisms identify significant environmental hazards while minimizing unnecessary alerts under different environmental conditions?

RQ6: What is the overall performance of the proposed system in terms of hazard detection, risk classification, risk map updating, and alert generation?

1.7 Scope limitations and assumptions

The scope of this research component is to develop an intelligent environmental monitoring system that identifies potential environmental hazards and represents their associated risk levels through a Dynamic Environmental Risk Map and Smart Safety Zones to support fall prevention among elderly people.

The proposed system focuses on monitoring selected environmental conditions, including insufficient lighting, wet or slippery floor conditions, nearby obstacles, stairs, and potentially hazardous furniture arrangements. IoT sensors, such as light and moisture sensors, are used to

collect environmental data. ESP32-CAM modules provide video streams of monitored areas, which are processed on a PC using a YOLO-based object detection model to identify relevant objects and potential hazards.

The collected sensor readings and camera-based detection results are used to assess environmental risk levels and classify monitored areas into three categories: Safe, Caution, and Danger. The Dynamic Environmental Risk Map is designed to reflect changes in environmental conditions, while the warning mechanism provides appropriate visual or audible alerts when significant hazards are detected.

The system will be evaluated in selected test environments using relevant performance measures, including hazard detection performance, risk classification performance, risk map updating, and alert generation. This research component focuses specifically on environmental risk assessment and does not independently perform comprehensive medical diagnosis or guarantee the prevention of falls.

The proposed system has several limitations:

1. Limited hazard coverage: The system focuses on selected environmental hazards and may not detect every possible cause of falls, such as sudden medical conditions, dizziness, or unpredictable human behaviour.
2. Camera and lighting limitations: The performance of camera-based detection may be affected by poor image quality, low lighting, camera positioning, occlusion, and the distance between objects and the camera.
3. Sensor accuracy and placement: Light and moisture sensor readings may be affected by sensor calibration, installation location, environmental conditions, and the characteristics of the monitored surfaces.
4. AI model limitations: The accuracy of YOLO-based detection depends on the training dataset, the hazard classes included in the model, and the similarity between test environments and the images used during training.
5. Limited monitoring coverage: The system can assess only the areas covered by its installed sensors and cameras. Hazards outside the monitored field of view or sensor coverage may remain undetected.
6. Network and processing dependency: Real-time performance may be affected by Wi-Fi connectivity, video transmission delays, PC processing capacity, and communication failures between system components.

7. Risk assessment constraints: The assigned risk levels depend on the selected rules, sensor thresholds, and detection results. Therefore, the generated risk map represents an estimated environmental risk level rather than a guaranteed prediction of whether a fall will occur.
8. Testing environment: Evaluation results obtained from selected laboratory or indoor test environments may not fully represent all real-world residential environments.

The proposed system is developed based on the following assumptions:

1. The installed IoT sensors are functional, appropriately calibrated, and positioned to collect relevant environmental measurements.
2. ESP32-CAM modules can provide usable video streams from the monitored areas under the available network and lighting conditions.
3. The PC running the YOLO-based object detection model has sufficient processing capacity to analyse the incoming video streams at a rate suitable for the intended monitoring application.
4. The selected YOLO model can detect the relevant objects and hazard-related classes included in its training or configuration, provided that the objects are sufficiently visible.
5. The selected sensor thresholds and risk assessment rules provide a reasonable basis for assigning environmental conditions to Safe, Caution, and Danger zones within the tested environment.
6. The monitored environment has adequate sensor and camera coverage to identify the selected hazards within the defined test areas.
7. Visual warnings and audible alerts can communicate detected hazards when the system, its communication network, and its alert mechanisms are functioning correctly

1.8 Individual contribution

My individual contribution to the research project, **Intelligent Fall Monitoring System for Elderly Care** is the design and development of the Environmental Risk Mapping and Smart Safety Zone System. The main objective of this component is to identify environmental hazards that may contribute to falls among elderly people and represent their associated risk levels dynamically in an understandable format.

My responsibilities include integrating IoT sensors, such as light sensors and moisture sensors, to monitor selected environmental conditions and configuring ESP32-CAM modules to obtain live video streams from monitored areas.

Another key contribution is integrating sensor readings with AI-based detection results to assess environmental risk levels. Based on predefined risk assessment rules, monitored areas are classified into three categories: Safe Zone, Caution Zone, and Danger Zone. Updating the Dynamic Environmental Risk Map according to changes in environmental conditions is also part of this work.

In addition, I am responsible for testing and evaluating the system while aiming to minimize unnecessary alerts. This evaluation considers performance measures such as hazard detection accuracy, risk classification performance, risk map updating, and alert generation.

Finally, I contribute environmental risk information that can be considered alongside the outputs of the other components of the overall research system. Through this component, I aim to support proactive fall prevention by helping elderly people and their caregivers identify potentially hazardous areas and make safer movement decisions.

1.9 Thesis structure

Chapter 2 reviews the literature on environmental fall risk, sensing, object detection and IoT messaging, and consolidates the research gap. Chapter 3 describes the research methodology. Chapter 4 presents the requirements and the design of the product. Chapter 5 describes the implementation, integration and testing. Chapter 6 reports the results, and Chapter 7 discusses them. Chapter 8 addresses commercialisation and sustainability, and Chapter 9 concludes the thesis and outlines future directions. The reference list, glossary and appendices follow.

2. LITERATURE REVIEW AND RELATED WORK

This chapter reviews the existing research, technologies, systems, and techniques relevant to the development of the Environmental Risk Mapping and Smart Safety Zone System for elderly fall prevention. It examines the theoretical foundations of environmental hazard monitoring, IoT-based sensing, computer vision, AI-based object detection, environmental risk assessment, and dynamic risk mapping.

The chapter also evaluates related research and existing solutions to identify their capabilities and limitations. Based on this review, the research gap is synthesized, and a conceptual framework is established to guide the design and development of the proposed system

2.1 Review method and scope

The literature review focuses on academic research and technical publications related to environmental hazards, elderly fall prevention, IoT-based monitoring, AI-based object detection, and intelligent risk assessment.

The review considers the following areas:

- Environmental hazards associated with falls among elderly people, including wet floors, insufficient lighting, obstacles, stairs, and unsafe furniture arrangements.
- IoT sensors used to monitor environmental conditions, particularly light and moisture.
- Camera-based monitoring and YOLO-based object detection for identifying relevant objects and potential hazards.
- Sensor data integration and rule-based environmental risk assessment.
- Dynamic risk maps, safety zone classification, and context-sensitive warning mechanisms.
- Existing fall prevention systems, environmental monitoring solutions, and related commercial products.

Search terms include elderly fall prevention, environmental hazard detection, IoT-based environmental monitoring, YOLO object detection, smart home safety, and dynamic risk mapping.

The review prioritizes relevant peer-reviewed publications and technical documentation. The selected studies are compared according to their objectives, technologies, detected hazards, risk assessment methods, visualization capabilities, and reported limitations.

The scope is limited to technologies and methods relevant to the proposed environmental monitoring component. Medical diagnosis, comprehensive physiological monitoring, and independent wearable-based fall detection are outside the primary scope of this individual component

2.2 Domain and theoretical foundations

2.2.1 Environmental Hazards and Elderly Fall Prevention

Environmental hazards can make it more difficult for elderly people to move safely within residential environments. Examples of such hazards include wet or slippery floors, insufficient lighting, obstacles along walking paths, stairs, uneven surfaces, and hazardous furniture arrangements.

Identifying these conditions before an accident occurs is an important aspect of preventing falls caused by environmental factors. Unlike systems that primarily focus on detecting a fall after it has occurred, environmental monitoring systems aim to identify conditions that may contribute to a fall in advance and communicate relevant risks in a timely manner.

Therefore, continuously monitoring environmental conditions and providing timely notifications about identified hazards can help create a safer living environment for elderly people and support the reduction of fall risks.

2.2.2 Internet of Things for Environmental Monitoring

The Internet of Things (IoT) enables physical sensors and embedded electronic devices to collect, exchange, and provide data for analysis and monitoring.

In the proposed system, IoT sensors are used to monitor environmental conditions. In particular, light sensors and moisture sensors are employed to measure ambient light levels and floor moisture conditions.

The measurements collected by these sensors can be transmitted to a processing application, where the data are analysed using predefined thresholds or risk assessment rules.

However, the overall safety of an environment cannot be determined solely from the readings of a single sensor. For example, a light sensor can identify insufficient lighting, but it cannot independently determine whether the floor is wet, obstacles are present, or stairs are located within the monitored area.

Therefore, integrating and analysing sensor data alongside other relevant environmental information is important for achieving a more comprehensive assessment of environmental safety.

2.2.3 Computer Vision and YOLO-Based Object Detection

Computer vision is a technology that enables computers to analyse images and video streams to identify and interpret visual information.

In the proposed system, ESP32-CAM modules are used to obtain video streams from monitored areas. These video streams are transmitted to a personal computer (PC), where they are analysed using a YOLO-based model. The system attempts to identify relevant objects and predefined hazard-related object classes within the monitored environment.

Therefore, the object detection results obtained from the YOLO model must be analysed alongside environmental sensor readings and the system's risk assessment rules. By combining these sources of information, the system can determine a more appropriate environmental risk level based on the detected conditions.

2.2.4 Dynamic Risk Mapping and Smart Safety Zones

A dynamic environmental risk map visually represents the estimated risk levels of monitored areas and updates when new information indicates a change in environmental conditions.

The proposed system uses three categories:

- Safe Zone: No significant environmental hazards are identified by the available monitoring mechanisms.
- Caution Zone: A potential hazard requires attention.
- Danger Zone: A more significant hazard requires an appropriate warning or preventive response.

This representation aims to help elderly people and caregivers understand environmental conditions and recognize potentially hazardous areas.

2.3 Related research

This section examines published research related to IoT-based elderly monitoring, environmental hazard detection, camera-based object detection, sensor fusion, and fall prevention.

The selected studies should be compared using the following criteria:

1. Research objective and application domain.
2. Sensors, cameras, and other technologies used.
3. Environmental hazards or events detected.
4. AI models and risk assessment methods.
5. Risk visualization and alert mechanisms.
6. Evaluation methods and reported performance.
7. Limitations and relevance to the proposed research.

The review should include research on IoT-based fall detection, AI-supported elderly safety, sensor-based fall-risk assessment, and computer vision in indoor environments. These studies provide background for identifying how environmental monitoring can complement systems that primarily focus on the individual.

Each study should be analysed critically rather than simply summarized. Particular attention should be given to whether it continuously monitors environmental hazards, combines different data sources, represents changing risk levels, and communicates risks before a fall occurs

2.4 Existing systems products and techniques

Existing fall prevention systems use wearable sensors, cameras, and IoT technologies to monitor elderly people and identify potential fall risks. Wearable devices detect body movements, while camera-based systems use computer vision to identify people and objects. However, these systems may not continuously monitor changing environmental hazards.

IoT technologies, such as ESP32 microcontrollers, light sensors, and moisture sensors, can monitor environmental conditions such as poor lighting and wet floors. AI-based object detection models, such as YOLO, can identify trained object classes from video streams captured by ESP32-CAM modules and processed on a computer. However, detection accuracy may be affected by lighting, camera positioning, and sensor calibration.Environmental risk mapping represents hazardous areas using categories such as Safe, Caution, and Danger. Combining sensor readings with AI-based visual detection can support dynamic risk

assessment and map updates. Visual indicators, buzzers, and mobile notifications can communicate significant risks. Therefore, the proposed system integrates IoT sensors, YOLO-based object detection, dynamic risk mapping, and appropriate alerts to help identify environmental hazards and support safer movement for elderly people.

Table 2.1: Comparison of representative approaches with the proposed component

Approach	Typical focus	Continuous ?	Environment-aware?	Spatial map?	Closed-loop alert?
Wearable fall detectors (pendants, smart watches)	Detect fall event / impact	Yes	No	No	Alert after fall
Camera-based fall detection	Detect person on floor / posture	Yes	Partly (scene only)	No	Alert after fall
Smart-home activity monitoring (PIR, door sensors)	Activity patterns, inactivity	Yes	No	Sometimes (room level)	Inactivity alerts
Manual home-hazard checklist (OT visit)	Lighting, clutter, flooring	No (periodic)	Yes	No	No
Proposed: Environmental Risk Mapping & Smart Safety Zones	Hazard quantification before a fall	Yes	Yes (light, distance, moisture, motion, vision)	Yes (dynamic 3D map)	Yes (local buzzer)

2.5 Gap synthesis and conceptual framework

Gap Synthesis

Existing fall prevention systems often focus on detecting human movements or falls, while continuous environmental hazard monitoring may be limited. Integrating live camera streams, IoT sensor readings, and AI-based object detection into a dynamic environmental risk assessment system remains an important area to investigate. Therefore, this research combines environmental sensor data with live video analysis to identify potential hazards and represent environmental risk levels through Smart Safety Zones.

Conceptual Framework

The proposed system uses ESP32-CAM modules to capture and stream live video from monitored areas to a computer. A YOLO-based AI model processes the incoming video frames on the computer to detect people and objects in real time. Meanwhile, light and moisture sensors collect environmental data. The video detection results and sensor readings are analysed together to estimate environmental risk levels. The system then displays live detection results and environmental risk information on a dashboard, representing monitored areas as Safe, Caution, or Danger zones and providing appropriate alerts when significant hazards are identified.

RESEARCH METHODOLOGY

2.6 Research design

This research follows an experimental system development approach to design and evaluate an Environmental Risk Mapping and Smart Safety Zone System for elderly care. The system integrates IoT sensor readings with AI-based object detection to identify potential environmental hazards. ESP32-CAM modules provide live video streams to a computer, where a YOLO model processes video frames to detect people and objects. Environmental sensor data and visual detection results are then used to assess risk levels and display relevant information on a monitoring dashboard.

2.7 Data participants or study materials

The study uses environmental sensor readings, live video streams, and object detection results as its primary data sources. Light and moisture sensors collect information about environmental conditions, while ESP32-CAM modules capture video from selected indoor areas. A YOLO model running on a computer detects relevant people and objects from the incoming video frames. The study materials include ESP32 hardware, sensors, cameras, a computer, the AI model, and the dashboard application. Testing can be conducted in controlled indoor environments using selected scenarios representing potential hazards. Human participants will only be involved if required for testing and after obtaining appropriate approval.

2.8 Experimental or study procedure

The experimental procedure begins by setting up the sensors and ESP32-CAM modules in selected monitoring areas. Sensor readings are collected, and live video is streamed to a computer. The YOLO model processes incoming video frames to identify configured object classes. The sensor readings and detection results are then analysed using predefined risk assessment rules. Based on the estimated risk, the system updates the environmental risk map and classifies monitored areas into Safe, Caution, or Danger zones. Finally, different test scenarios are conducted to evaluate hazard detection, risk classification, dashboard updates, and alert generation.

1. Power the sensor node and camera node and confirm that discovery messages are received and that both devices appear ONLINE.
2. Calibrate the dry-floor moisture baseline using the "Set dry baseline" control while the sensor is dry.

3. For each hazard scenario create the condition physically, wait for at least three telemetry cycles, and record the per-device score, zone score, level, and buzzer state.
4. Restore the neutral condition and confirm that scores and the buzzer return to the safe state (testing hysteresis).
5. Disconnect a device for longer than the offline timeout and confirm it is marked OFFLINE but remains listed.

2.9 Measures and evaluation metrics

Table 3.1: Evaluation measures

Measure	Definition	Target
Scoring correctness	Observed score equals the score expected from the rule table for the same inputs	100 % of tested scenarios
Level correctness	Observed LOW/MEDIUM/HIGH matches expected level	100 % of tested scenarios
Discovery success	Device appears in registry with correct type and zone without code changes	Both devices discovered automatically
Heartbeat accuracy	Device marked OFFLINE after the timeout and ONLINE again on the next message	Within 15 s ± one refresh cycle
Dashboard refresh latency	Time from message arrival to visible update	≈ 1 s (fragment refresh interval)
Vision throughput	YOLO frames per second reported by the worker	Reported; target stream 10 FPS
Alarm stability	No repeated toggling of the buzzer around the threshold	0 flapping events (hysteresis 30 %/20 %)

2.10 Data analysis

The data analysis combines the standard metrics with four checks of the data themselves, because the quality of the training data decides how far the model can be trusted.

1. **Metrics.** Accuracy, per-class precision, recall and F1-score, and the confusion matrix are computed from the held-out predictions. Feature importance is taken from the Random Forest.
2. **Label consistency.** The rule engine (Section 4.4) was applied to the synthetic dataset, using for each device its highest moisture reading as the dry baseline, and the resulting level was compared with the label.
3. **Sensor-axis and moisture-direction check.** The mean value of each sensor was compared across the three labels, and the direction of the moisture axis was compared with the behaviour of the real sensor (higher ADC value means drier).

4. **Domain-shift check.** The 21 real recordings were compared with the synthetic data in terms of sensor ranges and tilt angle, and the rule engine was applied to them.

The tools were Python, pandas, scikit-learn, matplotlib and seaborn. Error analysis considers low resolution and compression of ESP32-CAM frames, low light, occlusion, Wi-Fi delay and sensor noise as causes of errors

2.11 Ethics privacy security and responsible AI

The research followed the institute's ethical guidelines. [State whether ethical clearance was needed and obtained, and that volunteers gave informed consent.] No elderly participants were involved and no personal identifiers are stored by the component.

Privacy. The camera worker keeps only the latest annotated frame in memory; the code does not write video or images to disk, and frames are not sent to the cloud, because they travel from the camera to the PC on the local network. The cloud broker receives only sensor values and device information. The data-labelling tool stores numerical sensor values and a label, not images. Camera placement should avoid private areas such as bathrooms and bedrooms unless the occupant consents.

Security. Three weaknesses of the prototype were identified. First, the MQTT credentials are written in the firmware source, and the dashboard reads its own credentials from a .env file. Second, TLS certificate validation is disabled in the firmware (setInsecure) and in the dashboard (CERT_NONE and tls_insecure_set), so the broker identity is not verified. Third, the camera stream is plain HTTP without authentication. These are acceptable for a development prototype only; the recommended remedies are given in Section 5.5. For this reason the credentials are not reproduced in this thesis, and they should be rotated before the source code is published.

Responsible AI. The rule engine is transparent: every point of the score can be traced to one sensor value. The risk model is a decision-support tool; a false LOW may hide a hazard and a false HIGH causes unnecessary alerts, and neither replaces human supervision or a professional home assessment. YOLOv8n pre-trained on COCO may perform unevenly under different lighting, clothing, skin tone and postures, which has not been measured in this work. The system is a research prototype and not a certified medical device.

2.12 Reproducibility and research artefacts

The following artefacts support reproducibility: the firmware sources for the sensor node (SashiniESP32.ino) and the camera (CameraWebServer.ino with its board configuration

files); the dashboard (main.py); the data-labelling tool (data_labeller.py); the datasets (eldercare_synthetic_training_dataset.csv and eldercare_dataset.csv); the training script (train_model.py); and the trained artefacts (risk_classifier_rf.joblib, label_encoder.joblib, confusion_matrix.png and feature_importance.png). The random seed is 42 for the split, the forest and the cross-validation. [Provide the repository location, release tag and requirements.txt.

Two points must be resolved for exact reproduction. The training script requires a temperature column, which is not present in the synthetic CSV supplied with it, so the saved model (which has 14 features including temperature) cannot be reproduced from that file as it stands. In addition, the saved model was serialised with scikit-learn 1.9.0 and should be loaded with the same version. [Regenerate the dataset with the temperature column or retrain without it, and record the library version

3. PRODUCT AND SOLUTION DESIGN

This chapter presents the product designed to meet the research objectives: its requirements, its architecture within the complete solution, the alternatives that were considered and the detailed design of each part.

3.1 Requirements

Table 4.1: Functional requirements

ID	Requirement	Objective
FR1	Measure ambient light, distance, floor moisture, motion and inertial data at one reading per second.	O1
FR2	Devices announce themselves automatically (device id, type, zone, IP address, capabilities) without manual registration on the dashboard.	O2
FR3	Compute a 0–100 risk score and a LOW/MEDIUM/HIGH level per device and per zone.	O3
FR4	Show zones, devices and risk levels on a dynamic 3D map and in a zone summary.	O4
FR5	Allow the operator to switch the buzzer on and off remotely and activate it automatically when the floor is wet.	O4
FR6	Show live annotated camera video with YOLO detections for each camera device.	O5
FR7	Mark a device OFFLINE when no message has been received for a configured time, without removing it from the dashboard.	O2
FR8	Provide a tool to record labelled samples from live telemetry to a CSV file.	O6

Table 4.2: Non-functional requirements

ID	Requirement	Category
NFR1	Low cost: total prototype hardware cost suitable for home use.	Affordability
NFR2	Dashboard refresh of about one second; telemetry interval of one second.	Responsiveness
NFR3	Secure transport (TLS) and authenticated broker access.	Security
NFR4	Video is processed locally and is not stored.	Privacy
NFR5	Automatic recovery from WiFi, MQTT and camera-stream interruptions.	Reliability
NFR6	No hard-coded room names; zones are built from discovery messages.	Extensibility
NFR7	Risk scoring that is explainable to caregivers.	Transparency

3.2 Component architecture within the complete solution

Figure 4.1 shows the device and system architecture of the Environmental Risk Mapping and Smart Safety Zone System, arranged as eight units in two halves. The upper half is hardware. The sensing and edge unit is built around an ESP32 board that carries a BH1750 light sensor, a VL53L0X distance sensor, a moisture sensor, a PIR motion sensor, an MPU-6050 inertial sensor and a buzzer, powered by a TP4056 charger and an 18650 battery, with an ESP32-CAM as a separate camera node. The communication unit uses the built-in Wi-Fi of the ESP32 and the MQTT protocol over TLS to send data to the cloud server, a HiveMQ Cloud broker that receives and distributes the sensor data.

The lower half is software and AI processing on a PC. An MQTT client subscribes to the broker, discovers devices and zones automatically and sends buzzer commands back to the node. A rule-based risk engine converts light, distance, wetness and motion into a 0–100 score and a LOW, MEDIUM or HIGH level for each zone. A vision pipeline reads the MJPEG stream directly from the ESP32-CAM over Wi-Fi and runs YOLOv8n with OpenCV to detect people and objects. The live Streamlit dashboard presents the zone risk map, sensor values and annotated camera view to the caregiver. A Random Forest risk classifier is trained offline and is not yet part of the live decision path.

IOT Base Hardware & System Diagram

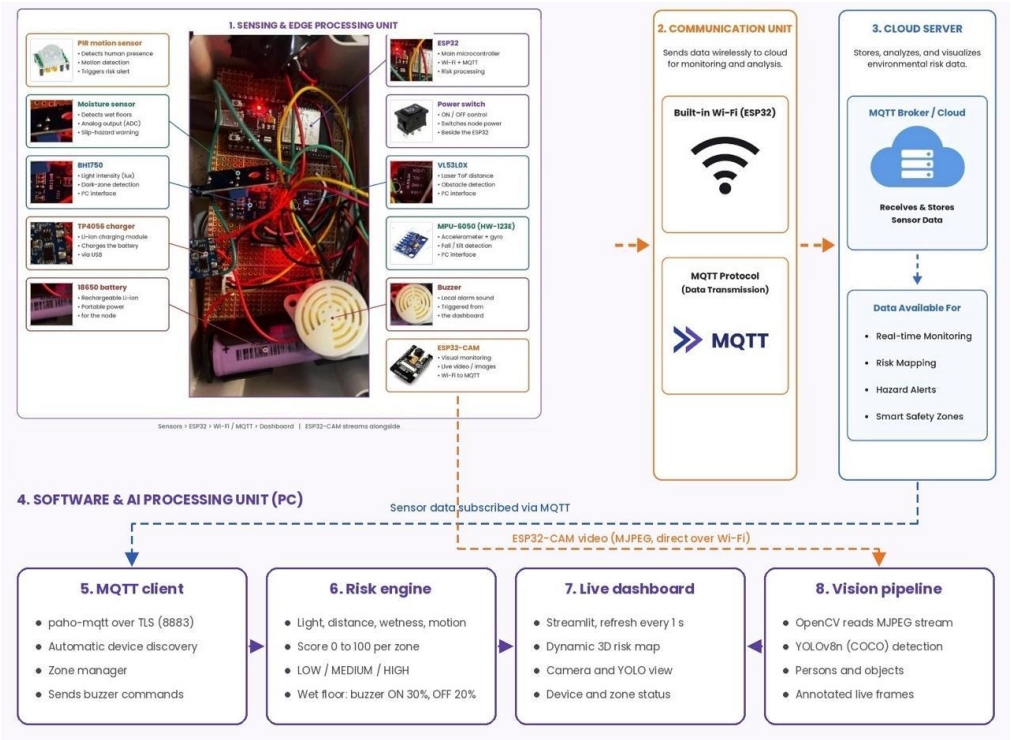


Figure 4.1: Architecture of the Environmental Risk Mapping & Smart Safety Zone System

Table 4.3: Layers of the architecture

Layer	Elements	Function
Perception	BH1750, VL53L0X, MPU6050, PIR, moisture sensor, ESP32-CAM	Measure light, distance, motion, presence, wetness and video
Edge	ESP32 sensor node, ESP32-CAM, buzzer	Sampling, Wi-Fi provisioning, MQTT publishing, command execution, MJPEG streaming
Messaging	HiveMQ Cloud MQTT broker (TLS, port 8883)	Publish/subscribe transport for discovery, status, telemetry and commands
Processing	Python on a PC: MQTT client, camera workers, YOLOv8n, risk engine, zone manager	Detection, risk scoring, zone aggregation, actuation
Presentation	Streamlit dashboard	Device cards, zone summary, 3D map, sensors, camera view, debug

3.3 Design alternatives and decisions

Table 4.4: Design alternatives and the decisions taken

Decision	Alternatives considered	Choice and reason
Messaging	HTTP polling; MQTT	MQTT: lightweight, publish/subscribe, retained discovery messages, bidirectional commands
Broker	Local Mosquitto; HiveMQ Cloud	HiveMQ Cloud: no server to maintain, TLS available, accessible from several components
Where to run YOLO	On the ESP32-CAM; on the host	Host: the ESP32 cannot run YOLO-class models in real time
Risk method	Rule-based; ML classifier	Both: rules are explainable and need no data; ML learns interactions and is evaluated offline
Dashboard	Custom web app; Streamlit	Streamlit: rapid development, native Plotly and image support, fragments for live refresh
Video format	RTSP; MJPEG over HTTP	MJPEG: supported by the ESP32 camera library and by OpenCV with no extra server
Zone model	Hard-coded rooms; discovered zones	Discovered zones: devices define their zone, so new rooms need no code change

3.4 Detailed design

3.4.1 Hardware design

The sensor node is built on an ESP32 development board. The BH1750, VL53L0X and MPU-6050 share the I2C bus (SDA on GPIO 21 and SCL on GPIO 22). The moisture sensor

analogue output is read on GPIO 34 (ADC), the PIR output on GPIO 27 and the buzzer is driven from GPIO 26. A power switch, an 18650 lithium-ion cell and a TP4056 charging module provide portable power. The ESP32-CAM is a separate device with an OV2640 camera that exposes an MJPEG stream.

Table 4.5: Hardware components and interfaces

Component	Measures or does	Interface
ESP32	Main controller, Wi-Fi, MQTT, edge processing	Wi-Fi; I2C (GPIO 21, 22)
BH1750	Light intensity (lux)	I2C
VL53L0X	Laser time-of-flight distance (mm)	I2C
MPU-6050	Acceleration, angular rate, temperature; tilt	I2C
Moisture sensor	Floor wetness (analogue)	ADC, GPIO 34
PIR sensor	Human presence	Digital input, GPIO 27
Buzzer	Local audible alarm	Digital output, GPIO 26
ESP32-CAM	Live video (MJPEG, QVGA)	Wi-Fi, HTTP port 81 (/stream)
18650 cell + TP4056	Portable power and USB charging	Power rail

3.4.2 Firmware and communication design

On start-up each device joins Wi-Fi, connects to the broker with TLS, publishes a retained-style discovery message that lists its identity, zone, IP, MAC, RSSI and sensors, and then publishes a status message. The sensor node publishes telemetry every 1,000 ms; the camera publishes a heartbeat status every 10,000 ms and a discovery message that includes its stream URL. The node also subscribes to its command topic and switches the buzzer when it receives the JSON command {"command":"buzzer","state":true} or false.

Table 4.6: MQTT topic structure

Topic	Direction	Content
eldercare/sensors/<id>/discovery	Node to PC	device_id, device_type, zone_id, ip, mac, rssi, sensors, actuators
eldercare/sensors/<id>/status	Node to PC	status (online/offline), zone_id, ip, rssi, buzzer
eldercare/sensors/<id>/telemetry	Node to PC	light, moisture, motion, distance, tof_timeout, accX..Z, gyroX..Z, temperature, buzzer
eldercare/sensors/<id>/command	PC to node	{"command":"buzzer","state":true|false}
eldercare/cameras/<id>/discovery	Camera to PC	device_type camera, zone_id, ip, stream_url, snapshot_url
eldercare/cameras/<id>/status	Camera to PC	Heartbeat (every 10 s)

3.4.3 Device and zone management

The PC application keeps a dictionary of devices and zones. A discovery message creates the device and, if it carries a zone identifier, creates the zone and adds the device to it. A device whose last message is older than 15 seconds is shown as OFFLINE; it is never deleted. This gives automatic discovery of devices and zones without manual configuration.

3.4.4 Risk engine

The risk engine converts the latest values of each sensor into an additive score as shown in Figure 4.2 and Table 4.6. The maximum is 30 + 35 + 30 + 5 = 100. A missing sensor reading contributes zero. The levels are LOW (0 to 30), MEDIUM (31 to 60) and HIGH (61 to 100). The zone score is the maximum score of its sensor nodes, and the zone colour on the map is determined by this score.

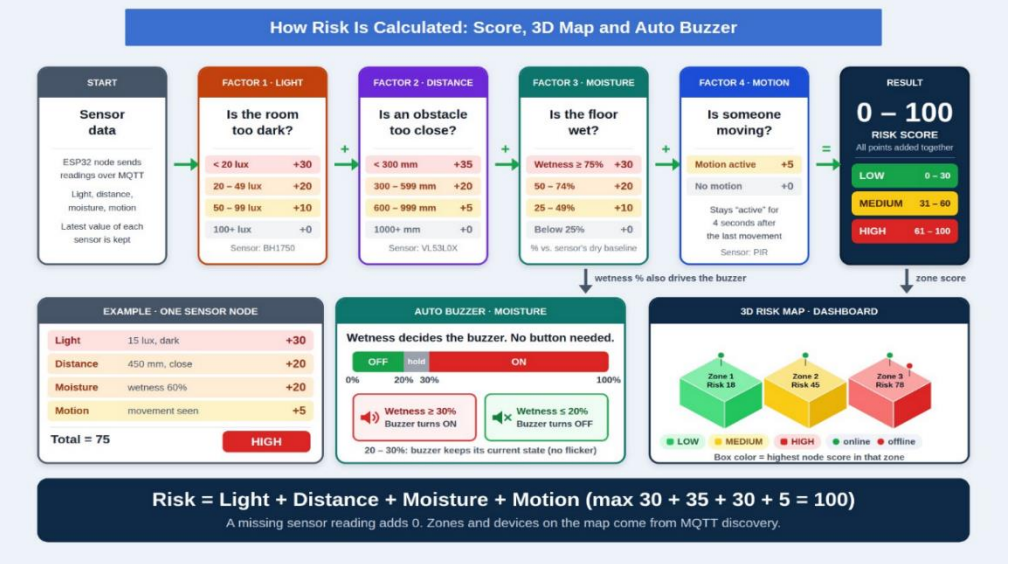


Figure 4.2: Risk calculation: four factors, score ranges, worked example, automatic buzzer and 3D map

3.4.5 Automatic buzzer with hysteresis

Wetness also drives the buzzer. When wetness reaches 30 % or more and the alarm is not already active, the PC publishes a buzzer ON command; when wetness falls to 20 % or less while the alarm is active, it publishes OFF. Between 20 % and 30 % the current state is kept. This hysteresis band prevents rapid on/off switching when the reading fluctuates around a single threshold, and no button press is required.

3.4.6 Vision pipeline

One worker thread per discovered camera opens the MJPEG URL with OpenCV, reads a frame, runs YOLOv8n, draws boxes, labels and confidence values, encodes the annotated frame as JPEG (quality 82) and stores it with the measured frame rate and detection count under a lock. The dashboard simply reads the latest snapshot. If the stream cannot be opened or read, the worker waits three seconds and reconnects. Because the worker is separate from the Streamlit refresh cycle, slow inference does not block the user interface.

3.5 Standards and professional practice

- MQTT 3.1.1 (OASIS standard) with TLS for transport security.
- I2C bus for the digital sensors; datasheet operating ranges of the sensor manufacturers were respected.
- Configuration through environment variables (.env) rather than hard-coded credentials.
- Thread-safe shared state (re-entrant locks) and daemon worker threads for robustness.

4. IMPLEMENTATION INTEGRATION AND TESTING

4.1 Development environment and technology stack

Table 5.1: Development environment and technology stack

Layer	Technology	Purpose
Sensors	BH1750, VL53L0X, MPU6050, PIR, analogue moisture probe	Light, distance, motion and tilt, presence, wetness
Microcontrollers	ESP32 (sensor node); ESP32-CAM AI-Thinker (camera); Arduino C++ firmware	Acquisition, Wi-Fi, MQTT, streaming
Provisioning	WiFiManager	Wi-Fi setup through a captive portal
Messaging	HiveMQ Cloud MQTT broker; PubSubClient on the devices; paho-mqtt (v2 API) on the PC	Discovery, status, telemetry, commands
Vision	OpenCV, Ultralytics YOLOv8n (yolov8n.pt)	Stream capture, detection, annotation
Dashboard	Streamlit with 1-second fragments; Plotly 3D (Mesh3d, Scatter3d)	User interface and risk map
Data and ML	pandas, scikit-learn (Random Forest), optional XGBoost, matplotlib, seaborn, joblib	Dataset handling, training, evaluation
Configuration	python-dotenv (.env file)	Broker credentials and YOLO parameters

4.2 Component implementation

4.2.1 Sensor-node firmware

The firmware initialises the I2C bus, the MPU-6050, BH1750 and VL53L0X, the PIR and moisture pins and the buzzer pin. The main loop keeps the Wi-Fi and MQTT connections alive, calls mqttClient.loop() and every 1,000 ms builds a compact JSON telemetry string and publishes it. The command callback normalises the incoming JSON and compares it with the compact strings for buzzer true and false, so the PC publishes JSON without spaces. A failed ToF read sets the tof_timeout flag.

4.2.2 Camera firmware

The camera firmware configures the OV2640, sets the frame size to QVGA for a stable stream, starts the HTTP server and publishes a discovery message containing http://<ip>:81/stream, followed by a heartbeat every 10 seconds.

4.2.3 Risk engine and buzzer logic

The core of the scoring function is shown below (abridged from main.py).

```
def calculate_risk(device):
    sensors = device.get("sensors", {})
    risk = 0
    light = sensors.get("light")
    if light is not None:
        risk += 30 if light < 20 else 20 if light < 50 else 10 if light < 100 else 0
    distance = sensors.get("distance")
    if distance is not None:
        risk += 35 if distance < 300 else 20 if distance < 600 else 5 if distance < 1000 else 0
    wetness = moisture_wetness_pct(device)
    if wetness is not None:
        risk += 30 if wetness >= 75 else 20 if wetness >= 50 else 10 if wetness >= 25 else 0
    if motion_active(device):
        risk += 5
    return min(risk, 100)
```

The buzzer hysteresis is implemented in the telemetry handler: when wetness is at least AUTO_BUZZER_ON_PCT (30) and the moisture_alarm flag is not set, the flag is set and control_buzzer(device_id, True) publishes the ON command; when wetness is at most AUTO_BUZZER_OFF_PCT (20) and the flag is set, the flag is cleared and the OFF command is published.

4.2.4 YOLO camera worker

The CameraWorker class owns the OpenCV capture, enforces a target of 10 frames per second by sleeping the remaining frame time, calls model.predict with conf 0.35, iou 0.45 and image size 640, and uses result.plot() to draw detections. The YOLO model is loaded lazily and only once, so the dashboard can still start and report a clear error if Ultralytics or the weights are missing.

4.2.5 Data labelling and training

data_labeller.py is a second Streamlit application that subscribes to the same topics and saves a CSV row with the selected label. train_model.py builds the derived features, performs the stratified split, five-fold cross-validation, trains the Random Forest (and XGBoost when installed), prints the classification reports and saves the model, the label encoder, the feature-importance plot and the confusion matrix.

4.3 Integration with the other components

The component integrates at three levels. At the data level, all information is available in the MQTT topics and in the dashboard state: the zone risk score and level, the device status, the YOLO detection count and the wetness and buzzer state. At the functional level, a HIGH zone risk or a wet floor is the natural input of the Smart Alert System, and the zone in which the

person is located can enrich the interpretation of events from the Activity Recognition and the PPG components. At the application level, the dashboard is one module of the group application. [Describe the interface actually implemented with each of the other components (for example the MQTT topics or the REST endpoints that they read) and what was tested.]

4.4 Verification and software testing

Testing combined several levels. Unit-level checks verified the rule engine against hand-calculated examples. Scenario tests exercised the whole chain by changing the conditions at the sensors and the camera and observing the dashboard. Integration tests checked discovery, status, telemetry and command messages between devices and dashboard. Performance checks measured frame rate and latency. Table 5.2 lists the planned test cases; the observed outcomes are reported in Chapter 6.

Table 5.2: Scenario and functional test cases

ID	Scenario	Expected outcome
TC1	Bright, dry, clear zone, nobody present	Score below 31: LOW (green)
TC2	Cover the light sensor (dark zone), everything else normal	Light adds 30 points; zone at least LOW–MEDIUM boundary as calculated
TC3	Place an object 25–50 cm in front of the ranging sensor	Distance adds 35 or 20 points
TC4	Wet the moisture probe to 25 %, 50 % and 75 %	Wetness adds 10, 20 and 30 points
TC5	Wetness rises to 30 %	Buzzer command published; buzzer on; status confirms
TC6	Wetness falls to 20 % or below	Buzzer off
TC7	Dark zone, obstacle close and wet floor together	Score at or near 100: HIGH (red)
TC8	Person walks in front of the camera	YOLO detects person; PIR motion YES for the hold time
TC9	Chair or table in front of the camera	YOLO detects the object class
TC10	Low-light scene	Detection quality recorded
TC11	Switch off a device or the Wi-Fi	Device marked OFFLINE after 15 s, kept on the dashboard, and online again on return
TC12	Interrupt the MQTT connection	Banner shows disconnection; reconnection restores data
TC13	Add a new device with a new zone	Zone and device appear without code change
TC14	Press the manual buzzer buttons	Command sent; state confirmed by the no

4.5 Deployment and operation

In operation, a device is powered on and, on first use, connected to Wi-Fi through its configuration portal. It then announces itself and appears on the dashboard. The caregiver starts the dashboard with Streamlit on the PC, which reads the broker settings and the YOLO parameters from the .env file, and views the zones, the sensors and the cameras. The sensor node is mounted [describe height and orientation], and the dry baseline of the moisture sensor is set while the floor is dry.

For a production deployment, five measures are recommended: validate the broker certificate on the devices and on the PC instead of disabling verification; give each device its own credentials and restrict it to its own topics; remove credentials from the source code and store them in secure configuration; protect the camera stream with authentication or put it on an isolated network; and require login for the dashboard. [State where the dashboard and broker run in the prototype.]

5. RESULTS AND VALIDATION

5.1 Research results

5.1.1 Dataset

Table 6.1: Class distribution of the synthetic training dataset (24,000 samples)

Class	Samples	Share
LOW	7,133	29.7 %
MEDIUM	9,598	40.0 %
HIGH	7,269	30.3 %

An 80 % / 20 % stratified split gave 19,200 training samples and 4,800 test samples. The test set contains 1,426 LOW, 1,920 MEDIUM and 1,454 HIGH samples.

5.1.2 Random Forest classification results

Figure 6.1 shows the confusion matrix of the Random Forest on the 4,800 test samples. Of these, 3,811 were classified correctly, giving an overall accuracy of 79.4 %. Table 6.2 gives the per-class metrics calculated from the confusion matrix.

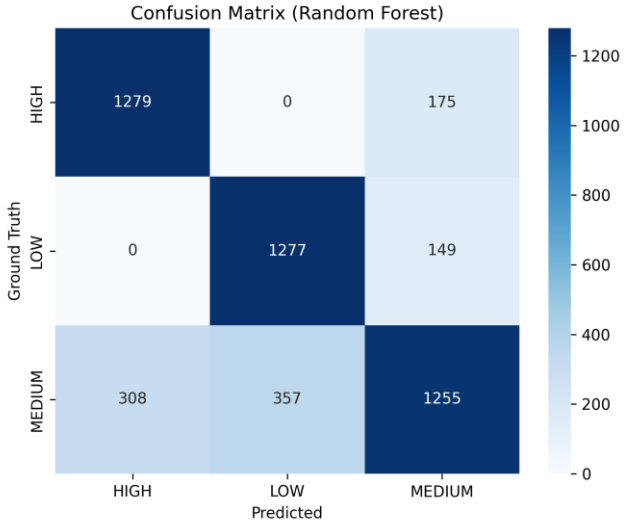


Figure 6.1: Confusion matrix of the Random Forest classifier on the test set

Table 6.2: Per-class performance of the Random Forest (test set, n = 4,800)

Class	Precision	Recall	F1-score	Support
HIGH	0.806	0.880	0.841	1,454
LOW	0.782	0.896	0.835	1,426
MEDIUM	0.795	0.654	0.717	1,920
Accuracy / macro F1	0.794	-	0.798	4,800

No LOW sample was predicted as HIGH and no HIGH sample as LOW. All errors involve the MEDIUM class: 175 HIGH and 149 LOW samples were predicted as MEDIUM, and 308 MEDIUM samples were predicted as HIGH and 357 as LOW.

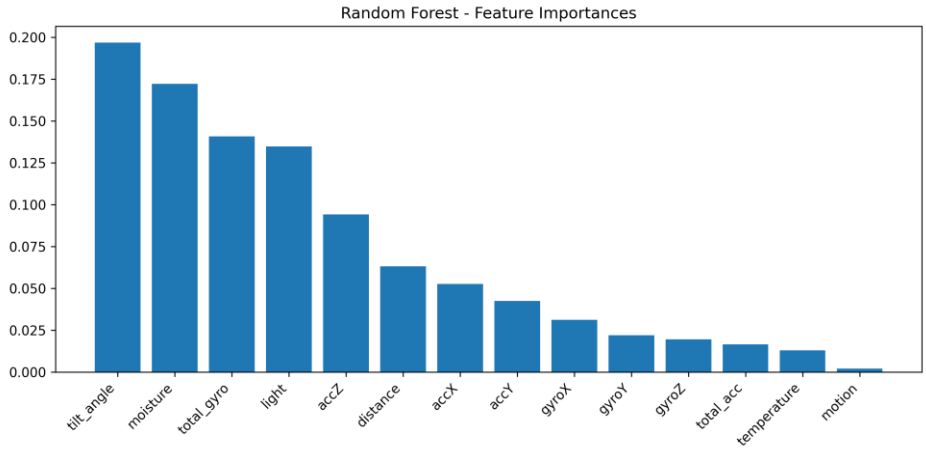


Figure 6.2: Random Forest feature importances

5.2 Product and technical results

5.2.1 Risk engine

All 18 offline test cases passed (Table 5.2). In the worked example of Figure 4.2 a node with 15 lux (+30), 450 mm distance (+20), 60 % wetness (+20) and detected motion (+5) scores 75 and is classified HIGH, which was reproduced exactly by the code (T16). The maximum score of 100 was reproduced by T17.

5.2.2 Live prototype

The prototype discovers devices automatically, displays device status, sensor cards, zone summaries, the 3D risk map and the annotated camera view on a single page that refreshes every second. Zone colours follow the highest node score, and a node that stops transmitting is shown OFFLINE after 15 seconds.

5.3 User or stakeholder validation

Describe the validation that was carried out: for example a demonstration to the supervisors, or feedback from potential caregivers or healthcare staff. If no formal user study was carried out, say so plainly and describe the informal feedback received (clarity of colours, usefulness of the 3D map and the buzzer)

5.4 Integration results

Report the outcome of integrating the component with the other components, for example that zone risk and wet-floor events are available to the Smart Alert System and that the dashboard is reachable from the group application.

5.5 Summary of findings

- The additive risk engine reproduces the specified scores for all tested inputs (18 of 18).
- A Random Forest reproduced the synthetic risk labels with 79.4 % accuracy and made no errors between the LOW and HIGH classes.
- MEDIUM is the hardest class (recall 65.4 %), because it lies between the other two.
- One implementation defect (missing-distance handling) was found by verification and has a one-line fix.

6. DISCUSSION

6.1 Interpretation of findings

The central design idea is that a living space can be treated as a set of zones, each described by a small set of physical measurements and by what the camera sees. Light, distance and wetness map directly onto recognised environmental causes of falls [4], [5], and the additive score makes the contribution of every hazard visible: a caregiver can see that a zone is red because it is dark and an obstacle is close, not because of an opaque model [22]. Taking the maximum over the nodes of a zone expresses that one hazard is enough to make a zone unsafe. The hysteresis of the buzzer is a practical lesson: without a gap between the on and off thresholds, a noisy analogue sensor would produce a stuttering alarm.

The machine-learning result must be read with care. An accuracy of 79.4 % and the absence of any HIGH/LOW confusion show that the three levels are separable in the training data and that the model's errors are conservative, since they stay between neighbouring levels. However, the model was trained on synthetic data whose labels disagree with the rule engine in half of the cases, whose moisture direction is opposite to the real sensor and whose inertial values differ from the real mounting. Consequently the accuracy describes how well the model learned the synthetic generator, not how well it predicts real environmental risk. The result is nevertheless useful: it shows that a pipeline for data labelling, training and evaluation exists, and that the data-analysis checks can find the problems before the model is deployed.

6.2 Comparison with literature baselines and existing solutions

A numerical comparison with published figures is not appropriate because datasets, sensors and conditions differ. In design terms, manual assessments are accurate but static, wearable detectors concern the person and not the place, and camera systems observe the scene but cannot measure illuminance or wetness. The present component combines continuous scalar sensing with vision and a zone-level, colour-coded output, at a low cost and with open, inspectable logic. Compared with mature commercial products, it lacks clinical validation, certification, a custom set of hazard classes and secure deployment.

6.3 Research contribution

a discovery-driven architecture in which devices announce their zone and capabilities over MQTT, so that zones, devices and camera workers are created without hard-coded rooms.

a transparent, additive environmental risk score for zones, with a maximum-over-nodes aggregation, and a hysteresis-based actuation rule for wet floors.

a working pipeline that streams ESP32-CAM video to a PC, runs YOLOv8n and shows annotated results alongside sensor data;

a data-labelling tool and a Random Forest baseline for environmental risk, with an honest analysis of the consistency, direction and realism of the training data.

6.4 Product contribution

A working IoT prototype with automatic discovery of devices and zones over MQTT, hysteresis-based wet-floor alarm, live YOLO video view and a dynamic 3D risk map.

A data-collection tool and training pipeline that allow the risk model to be improved with real data.

6.5 Limitations and threats to validity

Synthetic data. The classifier was trained and tested on synthetic samples; its 79.4 % accuracy shows learnability of the labelling scheme and must not be read as real-world accuracy. Only 21 real samples (20 LOW, 1 HIGH) from one node exist.

Threshold validity. Score weights and cut-offs are engineering choices, not clinically validated, and a floor-moisture sensor is a proxy for slipperiness.

Dry baseline. The dry reference is the highest ADC value seen so far; if a sensor starts wet, wetness is under-estimated until it dries.

Detector scope. YOLOv8n with COCO classes detects people and common objects but not hazards such as wet-floor patches, cables or rugs, and a QVGA ESP32-CAM image limits accuracy at distance and in poor light.

Rule-based production engine. The trained model is not yet integrated into the live dashboard, and the dashboard shows detections without fusing them into the risk score.

Security configuration. Certificate verification is disabled in the development setup.

Scale. Testing used one node, one camera and one zone.

6.6 Implications

The results suggest that an inexpensive sensor node plus a camera can give caregivers a continuously updated view of environmental hazards and can feed an environmental risk signal into personalised fall-prevention models. The main practical implication is that real data collection with several nodes and zones, and clinical review of the thresholds, are the next steps before field use.

7. COMMERCIALISATION AND SUSTAINABILITY

7.1 Value proposition and target users

The product gives older adults and their families a continuously updated, colour-coded picture of how safe each area of the home is, with an immediate local warning for wet floors. Primary users are older adults living at home and their family caregivers. Secondary users are elderly care homes, assisted-living facilities and home-care agencies that must monitor many rooms with limited staff.

7.2 Market and competitive position

Ageing populations increase demand for ageing-in-place technology. Existing products mostly focus on wearable fall detectors, emergency pendants or camera monitoring of the person. The competitive position of this component is environmental prevention at a low component cost, with an explainable score and local processing of video

7.3 Business or sustainability model

- Hardware kit: a sensor node plus optional ESP32-CAM sold or leased per room
- Subscription: dashboard, alerts, history and zone analytics for caregivers and care homes.
- Integration licence: environmental risk API for the personalised fall-prediction components of the full system.

7.4 Scalability, intellectual property and compliance

MQTT discovery allows new nodes and zones to be added without code changes, and the broker can serve many homes by using separate topic namespaces and accounts. The source code, the scoring scheme and the data-collection tooling are original work of the project team; third-party libraries (Streamlit, OpenCV, Ultralytics, paho-mqtt, scikit-learn and others) carry their own open-source licences, and Ultralytics YOLO is distributed under the AGPL-3.0 licence unless a commercial licence is purchased, which must be considered before commercial release. Compliance includes personal-data and privacy law of the deployment country (for Sri Lanka, the Personal Data Protection Act), informed consent for camera use, TLS with certificate validation, and, if marketed as a medical device, the relevant medical-device regulations.

7.5 Contribution to the Sustainable Development Goals

Table 8.1: Contribution to the Sustainable Development Goals [22]

Goal	Contribution
SDG 3: Good health and well-being	Prevents injuries by warning of hazards and supports healthy ageing at home
SDG 9: Industry, innovation and infrastructure	Low-cost IoT and AI innovation using open protocols and commodity hardware
SDG 11: Sustainable cities and communities	Safer, more inclusive homes and care facilities for older residents

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis presented the Environmental Risk Mapping and Smart Safety Zone System, an IoT and AI component of the Hybrid Multimodal Fall Prediction and Prevention System. The component senses light, obstacle distance, floor moisture, motion and inclination with an ESP32 node, streams video from an ESP32-CAM, transports data over secured MQTT, computes an explainable zone-level risk score with an automatic wet-floor buzzer, runs YOLOv8n on the camera stream and presents everything on a live dashboard with a dynamic 3D risk map. The rule engine reproduced all 18 designed test cases, and a Random Forest baseline reached 79.4 % accuracy on a synthetic test set without confusing LOW with HIGH. The prototype shows that continuous environmental risk mapping is feasible with low-cost hardware.

8.2 Achievement of objectives

Table 9.1: Achievement of objectives

Objective	Status	Evidence
1. Sensor node with secured MQTT	Achieved	Firmware; telemetry and discovery topics (Section 4.4.2)
2. Interpretable risk engine	Achieved	Tests T01 to T18; Figure 4.2
3. Auto discovery and automatic buzzer	Achieved	Section 4.4.3 and 4.4.5; functional tests
4. YOLO detection on ESP32-CAM stream	Achieved (pre-trained model)	Section 4.4.6; performance values to be entered in Table 6.3
5. Live dashboard and 3D risk map	Achieved	Section 4.4.7
6. Labelled dataset and ML baseline	Partly achieved	Synthetic data and 21 real samples; accuracy 79.4 %

8.3 Individual research and product contributions

The research contribution is the factor-based risk score and the evaluation of a learned baseline against it. The product contribution is the complete working prototype, including firmware, discovery-driven zone management, hysteresis alarm, vision pipeline, dashboard and data-collection tooling.

8.4 Future directions

- Collect real labelled data from several nodes and zones and retrain the classifier; integrate the trained model into the live dashboard.

- Fuse YOLO detections with the risk score, for example raising risk when a person is detected in a HIGH zone or when obstacles appear in walkways.
- Train a custom detector for hazards (wet-floor patches, cables, rugs, clutter) and consider fall and posture detection.
- Replace development TLS settings with certificate validation; add user authentication and encrypted storage.
- Add notifications to caregivers (mobile push or messaging) and a history view with trends.
- Clinical review of thresholds with occupational therapists and a field trial with older adults.
- Use the zone score to drive a guidance interface that directs users toward safer paths.

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GLOSSARY

Smart Safety Zone - An area of the home classified LOW, MEDIUM or HIGH risk from its sensor data.

Risk score - A 0 to 100 number obtained by adding the contributions of light, distance, moisture and motion.

Hysteresis - Using different ON and OFF thresholds so that an output does not switch rapidly around one threshold.

Dry baseline - The highest moisture ADC value observed from a sensor, used as 0 % wetness.

Discovery message - An MQTT message in which a device announces its identity, zone and capabilities.

MJPEG - Motion JPEG, a video stream made of consecutive JPEG images.

Time-of-flight - A ranging method that measures the travel time of emitted light.

APPENDICES

Appendix A Research instruments or extended results

A.1 Example messages. The following JSON examples correspond to the formats used by the firmware and the dashboard.

```
Topic: eldercare/sensors/NODE-003/telemetry
{"device_id":"NODE-003","zone_id":"stair_case","light":14.17,
 "moisture":4095,"motion":0,"distance":37,"tof_timeout":false,
 "accX":0.8721,"accY":0.5725,"accZ":0.0168,
 "gyroX":-0.871,"gyroY":-0.6543,"gyroZ":0.1304,"buzzer":0}

Topic: eldercare/sensors/NODE-003/command
{"command":"buzzer","state":true}
```

A.2 Columns of the collected dataset (eldercare_dataset.csv). timestamp, device_id, zone_id, temperature, light, moisture, motion, distance, tof_timeout, accX, accY, accZ, gyroX, gyroY, gyroZ, buzzer, label.

A.3 Derived features used by the classifier. total_acc = sqrt(accX^2 + accY^2 + accZ^2); total_gyro = sqrt(gyroX^2 + gyroY^2 + gyroZ^2); tilt_angle = arccos(accZ / total_acc) in degrees.

A.4 Random Forest settings. n_estimators 150, max_depth 12, min_samples_split 4, min_samples_leaf 2, random_state 42; 5-fold stratified cross-validation; 80 % / 20 % stratified split.

Appendix B Academic integrity and AI use disclosure

Table B.1: AI use disclosure

Item	Details
Tool	Anthropic Claude (large language model assistant)
Purpose	Drafting and structuring the thesis text from the author's project code, datasets and figures; computing metrics from the confusion matrix; executing offline unit tests of the risk functions
Representative workflow	The author supplied source code, datasets, figures and the thesis template; the assistant produced the draft document and the diagrams in Figure 4.2 and Figure 4.4
Verification performed	State what you checked: code behaviour, numbers, references, wording, and that all highlighted placeholders were replaced.
Sections or artefacts affected	Chapters 1 to 9, glossary and appendices; Figures 4.2 and 4.4